

Non-small cell lung cancer

Adenocarcinoma · squamous · large cell · **TNM 9th edition (Jan 2025)** · the driver landscape moves faster than any printed card — **re-check NCCN at every new diagnosis**

Compiled from the Stanford Oncology Handbook 2025, NSCLC (Salter A, Roy M) · NCCN · primary trial reports verified against PubMed, September 2026.

WORKUP

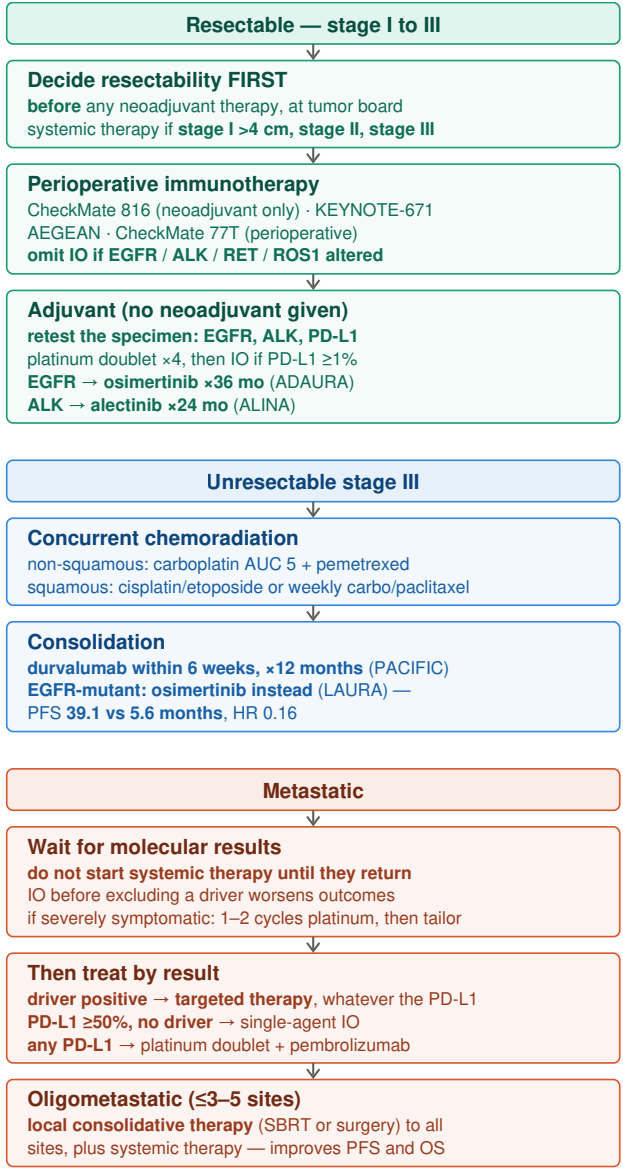
- **Chest CT with contrast** (nodes are far easier to call with contrast)
- **Brain MRI with contrast for every stage except IA** — PET never substitutes for it
- **PET/CT** for distant disease (or CT C/A/P + bone scan if unavailable)
- Biopsy: EBUS, CT-guided transthoracic, metastatic site, or pleural cytology. **Non-decalcified cores preferred for molecular**
- **Pathologic mediastinal nodal evaluation** for non-metastatic disease, especially stage II–III. EBUS reaches 2R/2L, 4R/4L, 7, 10, 11; **mediastinoscopy cannot reach stations 5–6** (needs a Chamberlain procedure)
- **PD-L1 in every case.** Molecular sequencing in all adenocarcinoma; add for squamous if never/light smoker, small biopsy, or adenosquamous
- **Stage IIIB or IV: concurrent tissue + liquid molecular testing** (ctDNA ~7-day turnaround). **A negative liquid biopsy does not rule out a driver** — confirm on tissue

STAGING — TNM 9TH (JAN 2025)

	Definition
T1 / T2	≤3 cm / >3–5 cm, or visceral pleura, adjacent lobe, or main bronchus
T3 / T4	>5–7 cm or chest wall, phrenic nerve, same-lobe nodule / >7 cm or mediastinum, great vessels, carina, esophagus, spine, different ipsilateral lobe
N1 / N2 / N3	Ipsilateral hilar (stations 10–14) / ipsilateral mediastinal or subcarinal — N2a single station, N2b multiple / contralateral or any supraclavicular
M1a / M1b	Pleural-pericardial effusion or nodules, or contralateral lung nodule / single extrathoracic met
M1c1 / M1c2	Multiple mets, single organ system / multiple organ systems

What changed from the 8th edition: N2 split into N2a/N2b and M1c split into M1c1/M1c2. **T descriptors are unchanged.**

TREATMENT PATHWAY



SCREENING

- **USPSTF 2021 (Grade B):** annual **low-dose CT**, age **50–80**, ≥20 pack-years, currently smoking or quit within 15 years
- Stop at **15 years since cessation**, or when comorbidity would preclude curative surgery
- **NLST:** 20% reduction in lung-cancer mortality vs chest radiography · **NELSON:** 24% in men
- **Smoking cessation remains the highest-yield intervention** — duration of smoking matters more than pack count

HISTOLOGY

Type	Features
Adenocarcinoma	~50%; peripheral, non/light smokers. TTF-1, napsin A, CK7 positive. Highest yield for actionable drivers. Mucinous (TTF-1–) is KRAS-enriched and EGFR-poor
Squamous	~25%; central, cavitating, heavy smokers. CK5/6, p40 / p63; keratin pearls and intercellular bridges. Pemetrexed and bevacizumab are both contraindicated
Large cell	Diagnosis of exclusion — treat as non-squamous
Adenosquamous	Both components, each ≥10% of the tumor

EARLY STAGE — SURGERY AND RADIATION

- **Lobectomy + mediastinal nodal sampling or dissection** is the standard; VATS preferred over thoracotomy
- **Sublobar resection is acceptable for peripheral cT1aN0 ≤2 cm**, after intraoperative nodal confirmation. **CALGB 140503:** non-inferior DFS (HR 1.01). **JCOG0802:** segmentectomy 5-yr OS 94.3% vs 91.1% (from fewer non-cancer deaths) **but double the local relapse** (10.5% vs 5.4%)
- **SBRT** for medically inoperable stage I. **Avoid central SBRT within 2 cm of the proximal bronchial tree**
- Preoperative assessment: **predicted postoperative FEV1 and DLCO**; <30% predicted flags high surgical risk

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DRIVER → THERAPY

Alteration	First line and beyond
EGFR ex19del / L858R <i>~15% US adeno</i>	Osimertinib 80 mg daily (FLAURA). Add carbo/pemetrexed ×4 for higher-risk disease — CNS involvement, L858R, TP53 co-mutation (FLAURA2: mPFS 25.5 vs 16.7 mo ; OS 47.5 vs 37.6). Or amivantamab + lazertinib (MARIPOSA: mPFS 23.7 vs 16.6 mo , HR 0.70; OS HR 0.75) — but grade ≥3 AEs 80% vs 52% , and it needs VTE prophylaxis for 4 months
EGFR exon 20 ins <i>~10% of EGFR</i>	Resistant to osimertinib and the classic TKIs. 1L amivantamab + carboplatin/pemetrexed (PAPILLON: mPFS 11.4 vs 6.7 mo , HR 0.40). Post-platinum: sunvozertinib (accelerated, 2L+ only). Mobocertinib withdrawn
ALK <i>~3–7%</i>	Lorlatinib 100 mg daily (CROWN: 5-yr PFS 60% vs 8% vs crizotinib, median not reached — the longest single-agent PFS reported in any solid tumor). Watch CNS effects (mood, cognition, seizures) and hyperlipidemia . Alternatives: alectinib, brigatinib, ensartinib
ROS1 <i>~1–2%</i>	Four co-preferred 1L agents: crizotinib, entrectinib, repotrectinib, taletrectinib. Prefer entrectinib / repotrectinib / taletrectinib if brain metastases — crizotinib has no CNS activity. Repotrectinib TKI-naïve ORR 79%, mPFS 35.7 mo. Alectinib has no ROS1 activity despite its ALK potency
RET <i>~1–2%</i>	Selpercatinib — the only randomized 1L evidence (LIBRETTO-431: mPFS 24.8 vs 11.2 mo vs chemo ± pembrolizumab, HR 0.47; intracranial ORR 82%). Pralsetinib is an alternative with more pneumonitis. Watch hypertension and transaminitis
MET exon 14 <i>~3–4%, older never-smokers</i>	Capmatinib (1L ORR 68%) or tepotinib (1L ORR ~57%) — single-arm, not comparable. Tepotinib: peripheral edema
BRAF V600E <i>~2%, smokers</i>	Encorafenib + binimetinib (PHAROS, treatment-naïve ORR 75%) or dabrafenib + trametinib (ORR 64%, mPFS 10.9 mo). Pyrexia is the classic combination toxicity — hold and dose-modify
KRAS G12C <i>~13% of adeno</i>	Not a first-line target — 1L is chemo-IO. 2L after platinum + anti-PD-(L)1: adagrasib (KRISTAL-12: mPFS 5.5 vs 3.8 mo, HR 0.58) or sotorasib (CodeBreak 200: 5.6 vs 4.5 mo, no OS gain; accelerated approval only)
HER2 (ERBB2) mutant <i>~2–3%, non-smoker women</i>	Mostly exon 20 insertions; high rate of brain metastases at presentation . Zongertinib — oral, and as of Feb 2026 it carries a 1L indication (treatment-naïve ORR 76%). T-DXd 5.4 mg/kg q3wk (DESTINY-Lung02: ORR 49%) — ILD in 13–15%, including fatal cases ; monitor with serial imaging
NTRK fusion <i><1%</i>	Larotrectinib, entrectinib or repotrectinib — all tumor-agnostic approvals
NRG1 fusion	Zenocutuzumab — accelerated approval, after progression on prior systemic therapy , for NSCLC and pancreatic adenocarcinoma only

PROGRESSION ON OSIMERTINIB

- **Rebiopsy** — resistance is heterogeneous, with no single dominant mechanism
- **On-target:** EGFR **C797S**. **Off-target:** MET amplification, HER2 amplification, acquired fusions
- **Histologic transformation is the one you cannot afford to miss: SCLC transformation** (with RB1 and TP53 loss) or squamous transformation. **Treat SCLC transformation as SCLC** — platinum/etoposide
- **Standard next step is platinum-doublet chemotherapy**
- **Amivantamab + chemotherapy** (MARIPOSA-2: mPFS 6.3 vs 4.2 mo, HR 0.48). **Approved regimen excludes lazertinib** (hematologic toxicity)
- **Datopotamab deruxtecan** — TROP2 ADC, accelerated approval, **EGFR-mutant only**

DRIVER-NEGATIVE METASTATIC

- **PD-L1 ≥50%: pembrolizumab** monotherapy (KEYNOTE-024: OS 30 vs 14 months); atezolizumab or cemiplimab are alternatives. **Use chemo-IO instead if the burden is high or the patient is symptomatic**
- **PD-L1 1–49% or <1%:** chemo-IO doublet. **Non-squamous** → pembrolizumab + carboplatin + pemetrexed (KEYNOTE-189). **Squamous** → pembrolizumab + carboplatin + paclitaxel (KEYNOTE-407)
- **Dual IO** — ipilimumab/nivolumab (CheckMate 227, 9LA) or durvalumab/tremelimumab (POSEIDON): chemo-sparing but more toxic, **most attractive at PD-L1 <1% and in squamous**
- **Second line: docetaxel 35 mg/m² days 1+8** — split dosing is better tolerated than 75 q3wk — ± ramucirumab

ADJUVANT THERAPY

- **Cisplatin doublet ×4** for stage II–IIIA (LACE: ~5% absolute 5-yr OS gain). **Stage IB only if ≥4 cm or high-risk features; no benefit and possible harm in stage IA**
- **EGFR → osimertinib ×36 months** (ADAURA: DFS HR 0.17, OS HR 0.49). **ALK → alectinib ×24 months** (ALINA: DFS HR 0.24). **Omit IO in both**
- **Atezolizumab** (IMpower010) is approved for **stage II–IIIA with PD-L1 ≥1%**, EGFR/ALK wild-type — OS NS overall; benefit concentrated at PD-L1 ≥50%
- **Pembrolizumab** (KEYNOTE-091) after adjuvant chemotherapy, **no PD-L1 restriction:** DFS HR 0.76 (PD-L1 ≥50% subgroup NS)

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PERIOPERATIVE AND ADJUVANT DOSING

Regimen	Dosing
CheckMate 816 <i>neoadjuvant only</i>	Cisplatin 75 mg/m² + pemetrexed 500 mg/m² (or gemcitabine 1250 d1+8) + nivolumab 360 mg on day 1 — q21d ×3 , then surgery
KEYNOTE-671 <i>perioperative</i>	Cisplatin 75 + pemetrexed 500 (or gemcitabine 1000 d1+8) + pembrolizumab 200 mg q21d ×4 → surgery → within 12 weeks, pembrolizumab 200 mg q21d ×13
AEGEAN <i>perioperative</i>	Cisplatin 75 + pemetrexed 500 (or gemcitabine 1250 d1+8) + durvalumab 1500 mg q21d ×4 → surgery → durvalumab 1500 mg q28d ×12
CheckMate 77T <i>perioperative</i>	Cisplatin 75 + pemetrexed 500 (or docetaxel 75) + nivolumab 360 mg q21d ×4 → surgery → within 90 days, nivolumab 360 mg q28d ×1 year
Adjuvant chemo	Cisplatin 75 mg/m² + pemetrexed 500 (non-squamous), gemcitabine 1250 d1+8, or docetaxel 75 — q21d ×4 . Growth factor support if using docetaxel
Adjuvant IO	Atezolizumab 1680 mg q4wk or pembrolizumab 200 mg q3wk , after adjuvant chemotherapy

CHEMORADIATION AND METASTATIC DOSING

Regimen	Dosing
ChemoRT non-squamous	Carboplatin AUC 5 + pemetrexed 500 mg/m² q21d ×4, with daily RT
ChemoRT squamous	Cisplatin 50 mg/m² days 1, 8, 29, 36 + etoposide 50 mg/m² days 1–5 and 29–33 ; or carboplatin AUC 2 + paclitaxel 45 mg/m² weekly — with daily RT
Consolidation durvalumab	10 mg/kg q14d ×12 months (or 1500 mg q4wk). Start within 6 weeks of finishing chemoRT
Platinum doublet + IO	Carboplatin AUC 5–6 + pemetrexed 500 (non-squamous) or paclitaxel 200 (squamous) + pembrolizumab 200 mg q21d ×4 → pemetrexed + pembrolizumab maintenance
Single-agent IO	Pembrolizumab 200 mg q3wk (or 400 q6wk) · atezolizumab 1200 mg q3wk · cemiplimab 350 mg q3wk
Osimertinib + chemo <i>FLAURA2</i>	Carboplatin AUC 6 + pemetrexed 500 q21d ×4, with osimertinib 80 mg daily continuously , then pemetrexed maintenance
Amivantamab + lazertinib	Amivantamab weight-based (<80 kg 1050 mg , ≥80 kg 1400 mg) — split over days 1–2 in week 1, weekly through week 5, then q2wk from week 7 ; + lazertinib 240 mg PO daily . VTE prophylaxis ×4 months
Docetaxel, 2L	35 mg/m² days 1+8 q21d — preferred over 75 q21d; reduce to 30 or 25 if heavily pretreated. ± ramucirumab 10 mg/kg
Pemetrexed maintenance	500 mg/m² q21d — non-squamous only

MONITORING AND CAUTIONS

- **Baseline HBsAg + anti-HBc before any immunosuppressive therapy**; consider quantitative TB testing by risk
- **Pemetrexed**: folic acid from ≥1 week before until 3 weeks after · B12 1000 mcg q9wk · dexamethasone 4 mg BID the day before and after, for rash. **Never in squamous**
- **Bevacizumab**: contraindicated in **squamous** histology, significant hemoptysis, and large cavitary tumors. Urine protein every other cycle; treat hypertension
- **Cisplatin**: avoid with hearing loss, CrCl <30, severe neuropathy, or poor performance status. Extensive hydration and aprepitant-containing antiemetics
- **Osimertinib**: echo at baseline and q3 months ×6 months; ECG at baseline, 2 weeks, 6 weeks, and with any QT-prolonging drug
- **Alectinib**: CK at baseline, 2 weeks, then q3 months · LFTs q2 weeks ×3 months. **Lorlatinib**: CNS effects and hyperlipidemia — median onset **15 days**; rosuvastatin first-line
- **Immunotherapy**: TSH and free T4 at baseline then q4–6 weeks. Prednisone **1–1.5 mg/kg** for irAEs with a slow taper
- **Do not start a targeted agent — osimertinib and crizotinib above all — within 3 months of immunotherapy**; the toxicity is additive. IO is also less effective in EGFR/ALK-driven disease

KEY TRIALS

CheckMate 816	Neoadjuvant platinum + nivolumab ×3 — pCR 24% vs 2%, EFS HR 0.63, OS HR 0.62
KEYNOTE-671	Perioperative pembrolizumab — EFS HR 0.58; the first perioperative IO trial with a confirmed OS benefit
AEGEAN / 77T	Perioperative durvalumab (EFS HR 0.68) / nivolumab (EFS HR 0.58)
ADAURA / ALINA	Adjuvant osimertinib ×36 mo (DFS HR 0.17, OS HR 0.49) / alectinib ×24 mo (DFS HR 0.24)
PACIFIC	Consolidation durvalumab after chemoRT — 5-yr OS 43% vs 33%
LAURA	Consolidation osimertinib, EGFR-mutant stage III — PFS 39.1 vs 5.6 mo , HR 0.16
FLAURA / FLAURA2	Osimertinib 1L; adding chemo gives mPFS 25.5 vs 16.7 mo and OS 47.5 vs 37.6
MARIPOSA	Amivantamab + lazertinib vs osimertinib — mPFS 23.7 vs 16.6 mo, OS HR 0.75 , at a real toxicity cost
CROWN	Lorlatinib 1L in ALK — 5-yr PFS 60% vs 8%
PAPILLON	Amivantamab + chemo 1L for EGFR exon 20 — mPFS 11.4 vs 6.7 mo
LIBRETTO-431	Selpercatinib vs chemo ± pembrolizumab in RET — mPFS 24.8 vs 11.2 mo
CALGB 140503	Sublobar resection non-inferior for peripheral cT1aN0 ≤2 cm — DFS HR 1.01